

Home Work 1 Solutions

II B. Tech. (Common)

Mathematics-II (NMCI102)

1. Solve the following systems of linear equations (if possible).

(a) $x_1 + x_2 = 4$, $x_2 - x_3 = 1$, $2x_1 + x_2 + 4x_3 = 7$ (b) $x_1 + 3x_2 + x_3 = 0$, $2x_1 - x_2 + x_3 = 0$

(c) $x_1 + 2x_2 - x_3 = 10$, $-x_1 + x_2 + 2x_3 = 2$, $2x_1 + x_2 - 3x_3 = 2$

(d) $x + y + z - 3w = 1$, $2x + 4y + 3z + w = 3$, $3x + 6y + 4z - 2w = 4$

(e) $x_1 + 2x_2 - x_3 = 10$, $-x_1 + x_2 + 2x_3 = 2$, $2x_1 + x_2 - 3x_3 = 8$.

(f) $x + y + z + 2w = 1$, $x + 2y + z + w = 2$.

Answers: (a) $(3, 1, 0)$ (b) $t(-4/7, -1/7, 1)$, $t \in \mathbb{R}$, (c) inconsistent (d) $(0, 0, 1, 0) + t(10, 0, -7, 1)$, $t \in \mathbb{R}$ (e) $(2, 4, 0) + t(5/3, -1/3, 1)$, $t \in \mathbb{R}$ (f) $(0, 1, 0, 0) + s(-1, 0, 1, 0) + t(-3, 1, 0, 1)$, $s, t \in \mathbb{R}$.

Solutions: (a) We solve the system using Gaussian elimination.

$$x_1 + x_2 = 4, x_2 - x_3 = 1, 2x_1 + x_2 + 4x_3 = 7.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & -1 & 1 \\ 2 & 1 & 4 & 7 \end{array} \right].$$

Apply the row operation

$$R_3 \rightarrow R_3 - 2R_1.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 4 & -1 \end{array} \right].$$

Next,

$$R_3 \rightarrow R_3 + R_2.$$

This gives the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 3 & 0 \end{array} \right].$$

From the third row,

$$x_3 = 0.$$

From the second row,

$$x_2 - x_3 = 1 \Rightarrow x_2 = 1.$$

From the first row,

$$x_1 + x_2 = 4 \Rightarrow x_1 = 3.$$

$$\boxed{(x_1, x_2, x_3) = (3, 1, 0)}$$

(b) We solve the system using Gaussian elimination.

$$x_1 + 3x_2 + x_3 = 0, 2x_1 - x_2 + x_3 = 0.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 2 & -1 & 1 & 0 \end{array} \right].$$

Apply the row operation

$$R_2 \rightarrow R_2 - 2R_1.$$

This gives the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -7 & -1 & 0 \end{array} \right].$$

From the second row,

$$-7x_2 - x_3 = 0 \Rightarrow x_2 = -\frac{1}{7}x_3.$$

Substituting into the first row,

$$x_1 + 3x_2 + x_3 = 0 \Rightarrow x_1 = -\frac{4}{7}x_3.$$

Let $x_3 = t$. The solution is

$$(x_1, x_2, x_3) = t \left(-\frac{4}{7}, -\frac{1}{7}, 1 \right), \quad t \in \mathbb{R}.$$

(c) We solve the system using Gaussian elimination.

$$x_1 + 2x_2 - x_3 = 10, \quad -x_1 + x_2 + 2x_3 = 2, \quad 2x_1 + x_2 - 3x_3 = 2.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 2 \end{array} \right].$$

Apply the row operations

$$R_2 \rightarrow R_2 + R_1, \quad R_3 \rightarrow R_3 - 2R_1.$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & -3 & -1 & -18 \end{array} \right].$$

Next,

$$R_3 \rightarrow R_3 + R_2.$$

This gives the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & -6 \end{array} \right].$$

Since the last row gives $0 = -6$, the system is inconsistent.

No solution

(d) We solve the system using Gaussian elimination.

$$x + y + z - 3w = 1, \quad 2x + 4y + 3z + w = 3, \quad 3x + 6y + 4z - 2w = 4$$

The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 6 & 4 & -2 & 4 \end{array} \right]$$

Apply the row operations

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 0 & 2 & 1 & 7 & 1 \\ 0 & 3 & 1 & 7 & 1 \end{array} \right]$$

Next, eliminate below the second pivot:

$$R_3 \rightarrow R_3 - \frac{3}{2}R_2$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 0 & 2 & 1 & 7 & 1 \\ 0 & 0 & -\frac{1}{2} & -\frac{7}{2} & -\frac{1}{2} \end{array} \right]$$

From the third row,

$$-\frac{1}{2}z - \frac{7}{2}w = -\frac{1}{2} \Rightarrow z = 1 - 7w$$

From the second row,

$$2y + z + 7w = 1 \Rightarrow y = 0$$

From the first row,

$$x + y + z - 3w = 1 \Rightarrow x = 10w$$

Let $w = t$. The solution set is

$$\boxed{(x, y, z, w) = (10t, 0, 1 - 7t, t), \quad t \in \mathbb{R}}$$

(e) We solve the system using Gaussian elimination.

$$x_1 + 2x_2 - x_3 = 10, \quad -x_1 + x_2 + 2x_3 = 2, \quad 2x_1 + x_2 - 3x_3 = 8$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 8 \end{array} \right]$$

Apply the row operations

$$R_2 \rightarrow R_2 + R_1, \quad R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & -3 & -1 & -12 \end{array} \right]$$

Next, eliminate below the second pivot:

$$R_3 \rightarrow R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

From the second row,

$$3x_2 + x_3 = 12 \Rightarrow x_2 = 4 - \frac{1}{3}x_3$$

From the first row,

$$x_1 + 2x_2 - x_3 = 10 \Rightarrow x_1 + 2\left(4 - \frac{1}{3}x_3\right) - x_3 = 10$$

$$x_1 + 8 - \frac{2}{3}x_3 - x_3 = 10 \Rightarrow x_1 - \frac{5}{3}x_3 = 2 \Rightarrow x_1 = 2 + \frac{5}{3}x_3$$

Let $x_3 = t$. The solution set is

$$(x_1, x_2, x_3) = \left(2 + \frac{5}{3}t, 4 - \frac{1}{3}t, t \right), \quad t \in \mathbb{R}$$

(f) We solve the system using Gaussian elimination.

$$x + y + z + 2w = 1, x + 2y + z + w = 2$$

The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 & 2 \end{array} \right]$$

Apply the row operation

$$R_2 \rightarrow R_2 - R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & -1 & 1 \end{array} \right]$$

From the second row,

$$y - w = 1 \Rightarrow y = 1 + w$$

From the first row,

$$x + y + z + 2w = 1 \Rightarrow x + (1 + w) + z + 2w = 1$$

$$x + z + 3w + 1 = 1 \Rightarrow x + z + 3w = 0 \Rightarrow x = -z - 3w$$

Let $z = s, w = t$. The solution set is

$$(x, y, z, w) = (-s - 3t, 1 + t, s, t), \quad s, t \in \mathbb{R}$$

2. Determine the conditions for which the following system $x + y + z = 1, x + 2y - z = b, 5x + 7y + az = b^2$ admits (a) only one solution (b) no solution (c) many solutions.

Answers: (a) $a \neq 1$ (b) $a = 1, b \neq -1, 3$ (c) $a = 1, b = -1$ or $b = 3$.

Solution: We consider the system:

$$x + y + z = 1, x + 2y - z = b, 5x + 7y + az = b^2$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & b \\ 5 & 7 & a & b^2 \end{array} \right], \quad A = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & a \end{array} \right]$$

Apply row operations:

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - 5R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & b - 1 \\ 0 & 2 & a - 5 & b^2 - 5 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & b-1 \\ 0 & 0 & a-1 & b^2-2b-3 \end{array} \right]$$

Rank analysis:

- If $a - 1 \neq 0$, then rank of $A = \text{rank of } [A|B] = 3 \implies$ unique solution.
- If $a - 1 = 0$ and $b^2 - 2b - 3 \neq 0$, then rank of $A = 2 < \text{rank of } [A|B] = 3 \implies$ no solution.
- If $a - 1 = 0$ and $b^2 - 2b - 3 = 0$, then rank of $A = \text{rank of } [A|B] = 2 < 3 \implies$ infinitely many solutions.

Conditions:

- Only one solution: $a \neq 1$
- No solution: $a = 1, b \neq -1, 3$
- Many solutions: $a = 1, b = -1$ or $b = 3$.

3. Consider the system:

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu,$$

where $\lambda, \mu \in \mathbb{R}$. Determine the conditions on λ and μ for which the system admits (a) only one solution (b) no solution (c) many solutions.

Solution : We have

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right], \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}$$

Apply row operations:

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array} \right]$$

Rank analysis:

- If $\lambda - 3 \neq 0$, then rank of $A = \text{rank of } [A|B] = 3 \implies$ unique solution.
- If $\lambda - 3 = 0$ and $\mu - 10 \neq 0$, then rank of $A = 2 < \text{rank of } [A|B] = 3 \implies$ no solution.
- If $\lambda - 3 = 0$ and $\mu - 10 = 0$, then rank of $A = \text{rank of } [A|B] = 2 < 3 \implies$ infinitely many solutions.

Conditions:

- Only one solution: $\lambda \neq 3$
- No solution: $\lambda = 3, \mu \neq 10$
- Many solutions: $\lambda = 3, \mu = 10$.

4. Using the row echelon form of a matrix, find the rank of the following matrices. (a) $A = \begin{bmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{bmatrix}$ (b)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 0 \end{bmatrix} \quad (c) \quad A = \begin{bmatrix} -2 & 0 & 0 & 3 \\ 1 & 5 & 3 & 0 \\ 3 & 2 & 1 & 6 \\ 3 & 5 & 3 & -3 \end{bmatrix}.$$

Answers: (a) 2 (b) 2 (c) 3

Solutions : (a) Given

$$A = \begin{bmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{bmatrix}$$

Row echelon form:

$$R_2 \rightarrow R_2 - \frac{3}{2}R_1, \quad R_3 \rightarrow R_3 - \frac{1}{2}R_1$$

$$\begin{bmatrix} 2 & 1 & 4 & 3 \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \end{bmatrix}, \quad R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 2 & 1 & 4 & 3 \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Number of non-zero rows: 2, thus, $\text{rank}(A) = 2$.

(b)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2.$$

(c)

$$A = \begin{bmatrix} -2 & 0 & 0 & 3 \\ 1 & 5 & 3 & 0 \\ 3 & 2 & 1 & 6 \\ 3 & 5 & 3 & -3 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow 2R_2 + R_1 \\ R_3 \rightarrow 2R_3 + 3R_1 \\ R_4 \rightarrow 2R_4 + 3R_1}} \begin{bmatrix} -2 & 0 & 0 & 3 \\ 0 & 10 & 6 & 3 \\ 0 & 4 & 2 & 21 \\ 0 & 10 & 6 & 3 \end{bmatrix} \xrightarrow{\substack{R_3 \rightarrow 10R_3 - 4R_2 \\ R_4 \rightarrow R_4 - R_2}} \begin{bmatrix} -2 & 0 & 0 & 3 \\ 0 & 10 & 6 & 3 \\ 0 & 0 & -4 & 198 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 3.$$

5. Find the rank of the matrix $A = \begin{bmatrix} a & -1 & 1 \\ -1 & a & -1 \\ -1 & -1 & a \\ 1 & 1 & 1 \end{bmatrix}$ if (a) $a \neq -1$ (b) $a = -1$. **Answers:** (a) 3 (b) 2.

Solution :

$$A = \begin{bmatrix} a & -1 & 1 \\ -1 & a & -1 \\ -1 & -1 & a \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{bmatrix} 1 & 1 & 1 \\ -1 & a & -1 \\ -1 & -1 & a \\ a & -1 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - aR_1}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & a+1 & 0 \\ 0 & 0 & a+1 \\ 0 & -1-a & 1-a \end{bmatrix}$$

$$\xrightarrow{R_4 \rightarrow R_4 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & a+1 & 0 \\ 0 & 0 & a+1 \\ 0 & 0 & 1-a \end{bmatrix}$$

$$\text{rank}(A) = \begin{cases} 3, & a \neq -1 \\ 2, & a = -1. \end{cases}$$

6. Find x such that the rank of the matrix $A = \begin{bmatrix} 1 & 3 & -3 & x \\ 2 & 2 & x & -4 \\ 1 & 1-x & 2x+1 & -5-3x \end{bmatrix}$ is 2. **Answers:** There does not exist any value of $x \in \mathbb{R}$.

Solution:

$$A = \begin{bmatrix} 1 & 3 & -3 & x \\ 2 & 2 & x & -4 \\ 1 & 1-x & 2x+1 & -5-3x \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \begin{bmatrix} 1 & 3 & -3 & x \\ 0 & -4 & x+6 & -4-2x \\ 0 & -2-x & 2x+4 & -5-4x \end{bmatrix}$$

$$\xrightarrow{R_3 \rightarrow (2+x)R_2 - 4R_3} \begin{bmatrix} 1 & 3 & -3 & x \\ 0 & -4 & x+6 & -4-2x \\ 0 & 0 & x^2-4 & -2x^2+8x+12 \end{bmatrix}$$

Since the third row is never zero for any real x , $\text{rank}(A) = 3 \quad \forall x \in \mathbb{R}$.

7. Find the value of a for which the rank of the matrix $\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & a & 1 & 0 \\ -1 & 1 & 1 & -1 \end{bmatrix}$ is 3.

Solution:

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & a & 1 & 0 \\ -1 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{R_4 \rightarrow R_4 + R_1} \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & a & 1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow 2R_3 - aR_2} \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & a+2 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_4 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & a+2 & 0 \end{bmatrix} \xrightarrow{R_4 \rightarrow R_4 - (a+2)R_3} \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & (a+2) \end{bmatrix}$$

For rank 3, the last row must be zero: $-(a+2) = 0 \implies a = -2$.

$$\boxed{a = -2.}$$

8. Consider the system: $3x - y + z = 1$, $2x + y - z = 2$, $x - 2y + 2z = 5$. Find the rank of the coefficient matrix and the augmented matrix. Hence, determine the consistency of this system.

Solution:

$$\text{Coefficient matrix: } A = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 1 & -1 \\ 1 & -2 & 2 \end{bmatrix}, \quad \text{Augmented matrix: } [A|b] = \begin{bmatrix} 3 & -1 & 1 & 1 \\ 2 & 1 & -1 & 2 \\ 1 & -2 & 2 & 5 \end{bmatrix}$$

Step 1: Row reduce the coefficient matrix:

$$R_2 \rightarrow 3R_2 - 2R_1, \quad R_3 \rightarrow 3R_3 - R_1$$

$$\begin{bmatrix} 3 & -1 & 1 \\ 0 & 5 & -5 \\ 0 & -5 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 3 & -1 & 1 \\ 0 & 5 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of coefficient matrix: $r(A) = 2$.

Step 2: Row reduce the augmented matrix:

$$R_2 \rightarrow 3R_2 - 2R_1, \quad R_3 \rightarrow 3R_3 - R_1$$

$$\begin{bmatrix} 3 & -1 & 1 & 1 \\ 0 & 5 & -5 & 4 \\ 0 & -5 & 5 & 14 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 3 & -1 & 1 & 1 \\ 0 & 5 & -5 & 4 \\ 0 & 0 & 0 & 18 \end{bmatrix}$$

Rank of augmented matrix: $r([A|b]) = 3$.

Since $r(A) \neq r([A|b])$, the system is inconsistent.

9. Let P be a non-zero matrix of order $m \times 1$ and Q be a non-zero matrix of order $1 \times n$. Determine $\text{rank}(PQ)$.

Solution: Since P is a non-zero column matrix,

$$\text{rank}(P) = 1.$$

Since Q is a non-zero row matrix,

$$\text{rank}(Q) = 1.$$

Using the rank inequality,

$$\text{rank}(PQ) \leq \min\{\text{rank}(P), \text{rank}(Q)\} = \min\{1, 1\} = 1.$$

But $PQ \neq 0$ (why?), hence

$$\text{rank}(PQ) \neq 0.$$

Therefore,

$$\boxed{\text{rank}(PQ) = 1.}$$

10. Let P be a matrix of order $m \times n$ and Q be a matrix of order $n \times p$, $n \neq p$. If $\text{rank}(P) = n$ and $\text{rank}(Q) = p$, determine $\text{rank}(PQ)$.

Solution: Since $\text{rank}(P) = n$, P has full column rank. Hence there exists an invertible $m \times m$ matrix E such that the row reduced echelon form of P is

$$EP = \begin{pmatrix} I_n \\ 0 \end{pmatrix}.$$

Now consider

$$EPQ = \begin{pmatrix} I_n \\ 0 \end{pmatrix} Q = \begin{pmatrix} Q \\ 0 \end{pmatrix}.$$

Since elementary row operations do not change rank,

$$\text{rank}(PQ) = \text{rank}(EPQ) = \text{rank} \begin{pmatrix} Q \\ 0 \end{pmatrix} = \text{rank}(Q) = p.$$

11. Using the row reduced echelon form of a matrix (Gauss-Jordan Method), find the inverse of the following matrices.

(a) $\begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{bmatrix}$. Answers: (a) $\begin{bmatrix} 1/8 & -5/8 & 3/4 \\ -1/4 & 3/4 & -1/2 \\ 3/8 & -3/8 & 1/4 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & -1/2 & -2 \\ -1 & 1/2 & 0 \\ -3 & 0 & 1 \end{bmatrix}$.

Solutions: (a) Start with the matrix $[A|I]$:

$$\left[\begin{array}{ccc|ccc} 0 & 2 & 4 & 1 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

Step 1: Swap $R_1 \leftrightarrow R_2$ to get a non-zero pivot in the first row:

$$\left[\begin{array}{ccc|ccc} 2 & 4 & 2 & 0 & 1 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

Step 2: Make pivot 1 in row1: $R_1 \rightarrow \frac{1}{2}R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1/2 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

Step 3: Apply $R_3 \rightarrow R_3 - 3R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1/2 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 0 & -3 & -2 & 0 & -3/2 & 1 \end{array} \right]$$

Step 4: Pivot in row2, column2: $R_2 \rightarrow \frac{1}{2}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1/2 & 0 \\ 0 & 1 & 2 & 1/2 & 0 & 0 \\ 0 & -3 & -2 & 0 & -3/2 & 1 \end{array} \right]$$

Step 5: Apply $R_1 \rightarrow R_1 - 2R_2$, $R_3 \rightarrow R_3 + 3R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & 1/2 & 0 \\ 0 & 1 & 2 & 1/2 & 0 & 0 \\ 0 & 0 & 4 & 3/2 & -3/2 & 1 \end{array} \right]$$

Step 6: Pivot in row3, column3: $R_3 \rightarrow \frac{1}{4}R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & 1/2 & 0 \\ 0 & 1 & 2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 3/8 & -3/8 & 1/4 \end{array} \right]$$

Step 7: Apply $R_1 \rightarrow R_1 + 3R_3$, $R_2 \rightarrow R_2 - 2R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/8 & -5/8 & 3/4 \\ 0 & 1 & 0 & -1/4 & 3/4 & -1/2 \\ 0 & 0 & 1 & 3/8 & -3/8 & 1/4 \end{array} \right]$$

Thus, the inverse is: $A^{-1} = \begin{bmatrix} 1/8 & -5/8 & 3/4 \\ -1/4 & 3/4 & -1/2 \\ 3/8 & -3/8 & 1/4 \end{bmatrix}$.

(b)

Step 1: Augment with the identity matrix

$$[A | I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 4 & 4 & 0 & 1 & 0 \\ 3 & 3 & 7 & 0 & 0 & 1 \end{array} \right]$$

Step 2: Eliminate entries below the pivot in column 1

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right]$$

Step 3: Make the pivot in row 2 equal to 1

$$R_2 \rightarrow \frac{1}{2}R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right]$$

Step 4: Eliminate entries above and below the pivot in column 2

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 2 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right]$$

Step 5: Eliminate entries above the pivot in column 3

$$R_1 \rightarrow R_1 - 2R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & -\frac{1}{2} & -2 \\ 0 & 1 & 0 & -1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right]$$

Step 6: The inverse is

$$A^{-1} = \begin{bmatrix} 8 & -\frac{1}{2} & -2 \\ -1 & \frac{1}{2} & 0 \\ -3 & 0 & 1 \end{bmatrix}.$$